

Robert Bogdan Staszewski

professor

基本信息

姓名: Robert Bogdan Staszewski 所属院校: 都柏林大学学院
地区: Ireland 年龄范围: 50岁以上
海外经历: 有 企业经历: 有

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

微电子科学与工程

通信工程

集成电路设计与集成系统

研究领域

射频集成电路

模拟与数模混合电路

模拟与射频IC设计

低功耗电路与能量管理

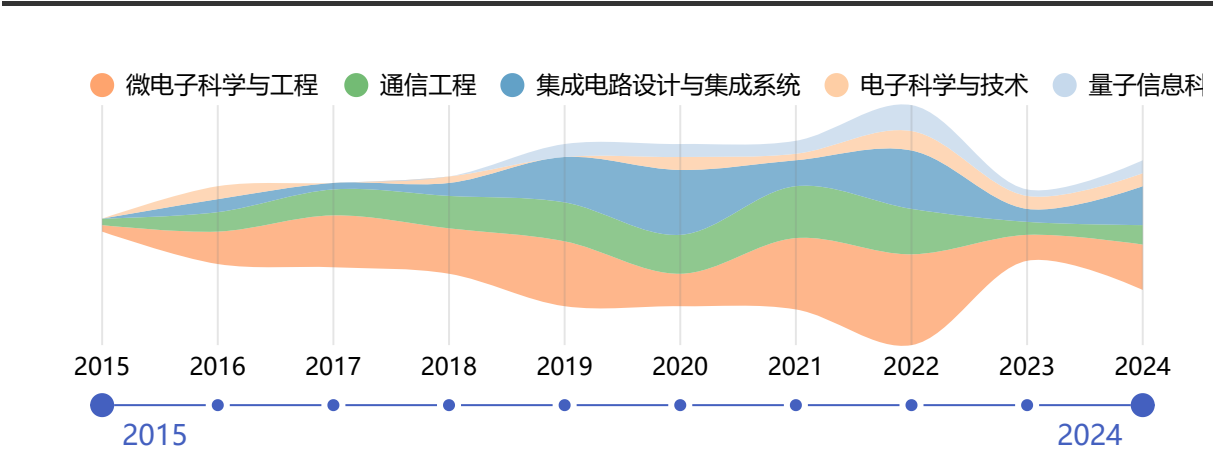
移动通信与5G/6G

个人简介

Robert Bogdan Staszewski is a highly accomplished academic and researcher. Before 2009, he was a design engineer and industrial researcher in microelectronics and communication systems at Texas Instruments. He invented the Digital RF Processor technology, resulting in numerous patents and publications. In 2009, he joined Delft University of Technology as a

tenured academic. Since 2014, he has been a Full Professor at University College Dublin, leading research in microelectronics for IoT and quantum computers. He co - founded Equal1 to build a CMOS quantum computer. He is an IEEE Fellow and won the IEEE Circuits and Systems Industrial Pioneer Award. He holds a Ph.D., M.Sc., and B.Sc. in Electrical Engineering from the University of Texas at Dallas.

近年发表的论文



A Fully Integrated QPSK/16-QAM D-Band CMOS Transceiver With Mixed-Signal Baseband Circuitry Realizing Digital Interfaces

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Guan, Pingda; Ma, Ruichang; Jia, Haikun; Deng, Wei; Deng, Mingxing; Xue, Jiamin; Yan, Angxiao; Sun, Shiyan; Peng, Qiuyu; Siriburanon, Teerachot; Staszewski, Robert Bogdan; Wang, Zhihua; Chi, Baoyong
被引用次数: 0 2024

Canceling Fundamental Fractional Spurs Due to Self-Interference in a Digital Phase-Locked Loop

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Gao, Zhong; Staszewski, Robert Bogdan; Babaie, Masoud
被引用次数: 1 2024

Charge-Based Single-Shot Readout Systems in Quantum Computing

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Power, Conor; Staszewski, Robert Bogdan; Blokhina, Elena
被引用次数: 0 2024

RFIC Reuse Techniques to Enable Ultra-Low-Power IoT: A Tutorial

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Xu, Kai; Kong, Xiangjian; Staszewski, Robert Bogdan
被引用次数: 0 2024

Monolithically Integrated Quantum Dots in a 22-nm Fully Depleted Silicon-on-Insulator Process Operating at 3 K

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Bashir, Imran; Sokolov, Andrii; Wu, Xutong; Giounanlis, Panagiotis; Petropoulos, Nikolaos; Leipold, Dirk; Asker, Mike; Esmailyan, Ali; Andrade-Miceli, Dennis; Haenlein, Hans-Christoph; Mcgeough, Conor; Staszewski, Robert Bogdan; Blokhina, Elena

被引用次数: 0

2024

A Low-Noise Digital-to-Time Converter Exploiting Waveform of Integrated Crystal Oscillator

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Siriburanon, Teerachot; Chen, Xi; Liu, Chunxiao; Du, Jianglin; Zhu, Anding; Staszewski, Robert Bogdan

被引用次数: 0

2024

A 150-GHz On-Chip Antenna with 40-GHz Bandwidth in 22-nm FD-SOI CMOS for Automotive Phased Array Radars

2024 IEEE INTERNATIONAL SYMPOSIUM ON ANTENNAS AND PROPAGATION AND INC/USNCURSI RADIO SCIENCE MEETING, AP-S/INC-USNC-URSI 2024

Bozorg, Amir; Sharbati, Vahid; Bao, Xiulong; Staszewski, Robert Bogdan

被引用次数: 0

2024

Millimeter-Wave All-Digital Phase-Locked Loop Using Reference Waveform Oversampling Techniques

IEEE OPEN JOURNAL OF THE SOLID-STATE CIRCUITS SOCIETY

Siriburanon, Teerachot; Liu, Chunxiao; Du, Jianglin; Staszewski, Robert Bogdan

被引用次数: 0

2024

Nonlinearity-Induced Spur Analysis in Fractional-N Synthesizers With $\Delta\Sigma$ Quantization Cancellation

IEEE OPEN JOURNAL OF THE SOLID-STATE CIRCUITS SOCIETY

Hu, Yizhe; Tao, Weichen; Staszewski, Robert Bogdan

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2024

A Two-Story Quad-Core Dual-Mode VCO in 65-nm CMOS

IEEE SOLID-STATE CIRCUITS LETTERS

Guan, Pingda; Jia, Haikun; Deng, Wei; Ma, Ruichang; Liao, Huabing; Siriburanon, Teerachot; Staszewski, Robert Bogdan; Wang, Zhihua; Chi, Baoyong

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2024

8-Shaped Inductors: An Essential Addition to RFIC Designers' Toolbox

IEEE OPEN JOURNAL OF THE SOLID-STATE CIRCUITS SOCIETY

Guan, Pingda; Jia, Haikun; Deng, Wei; Siriburanon, Teerachot; Staszewski, Robert Bogdan; Wang, Zhihua; Chi, Baoyong

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Exploring Speed Maximization of Frequency-to-Digital Conversion for Ultra-Low-Voltage VCO-Based ADCs

A Digital PLL-Based Phase Modulator With Non-Uniform Clock Compensation and Non-linearity Predistortion

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Gao, Zhong; Fritz, Martin; Spalink, Gerd; Staszewski, Robert Bogdan; Babaie, Masoud

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2023

A 30-GHz Class-F Quadrature DCO Using Phase Shifts Between Drain-Gate-Source for Low Flicker Phase Noise and I/Q Exactness

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Chen, Xi; Hu, Yizhe; Siriburanon, Teerachot; Du, Jianglin; Staszewski, Robert Bogdan; Zhu, Anding

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A Low-Spur Fractional-N PLL Based on a Time-Mode Arithmetic Unit

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Gao, Zhong; He, Jingchu; Fritz, Martin; Gong, Jiang; Shen, Yiyu; Zong, Zhirui; Chen, Peng; Spalink, Gerd; Eitel, Ben; Alavi, Morteza S.; Staszewski, Robert Bogdan; Babaie, Masoud

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An On-Chip Picoampere-Level Leakage Current Sensor for Quantum Processors in 22-nm FD-SOI CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Esmailian, Ali; Blokhina, Elena; Miceli, Dennis Andrade M.; Faust, Eoin; Giounanlis, Panagiotis; Leipold, Dirk; Wang, Hongying; Bashir, Imran; Koskin, Eugene; Siriburanon, Teerachot; Staszewski, Robert Bogdan

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2023

A Compact, Low-Power, Low-NF, Millimeter-Wave Cascode LNA With Magnetic Coupling Feedback in 22-nm FD-SOI CMOS for 5G Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kobal, Enis; Siriburanon, Teerachot; Staszewski, Robert Bogdan; Zhu, Anding

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2023

A Deep-Subthreshold Variation-Aware 0.2-V Open-Loop VCO-Based ADC

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Viet Nguyen; Schembari, Filippo; Staszewski, Robert Bogdan

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2022

Topological order detection and qubit encoding in Su-Schrieffer-Heeger type quantum dot arrays

JOURNAL OF APPLIED PHYSICS

Petropoulos, Nikolaos; Bogdan Staszewski, Robert; Leipold, Dirk; Blokhina, Elena

被引用次数: 6

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A Charge-Sharing IIR Filter With Linear Interpolation and High Stopband Rejection

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Bozorg, Amir; Staszewski, Robert Bogdan

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2022

A mm-Wave Switched-Capacitor RFDAC

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hieu Minh Nguyen; Walling, Jeffrey Sean; Zhu, Anding; Staszewski, Robert Bogdan

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A Fractional-N Digitally Intensive PLL Achieving 428-fs Jitter and <-54-dBc Spurs Under 50-mV ΔV_{DD} Supply Ripple

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Chen, Yue; Gong, Jiang; Staszewski, Robert Bogdan; Babaie, Masoud

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A Charge-Sharing Locking Technique With a General Phase Noise Theory of Injection Locking

IEEE JOURNAL OF SOLID-STATE CIRCUITS

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A Compact 0.2-0.3-V Inverse-Class-F23 Oscillator for Low $1/f_3$ Noise Over Wide Tuning Range

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Du, Jianglin; Hu, Yizhe; Siriburanon, Teerachot; Kobal, Enis; Quinlan, Philip; Zhu, Anding; Staszewski, Robert Bogdan

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A Clock-Phase Reuse Technique for Discrete-Time Bandpass Filters

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2022

A Gm-Boosting Technique for Millimeter-Wave Low-Noise Amplifiers in 28-nm Triple-Well Bulk CMOS Using Floating Resistor in Body Biasing

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kobal, Enis; Siriburanon, Teerachot; Chen, Xi; Hieu Minh Nguyen; Staszewski, Robert Bogdan; Zhu, Anding

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Flicker Phase-Noise Reduction Using Gate-Drain Phase Shift in Transformer-Based Oscillators

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Chen, Xi; Hu, Yizhe; Siriburanon, Teerachot; Du, Jianglin; Staszewski, Robert Bogdan; Zhu, Anding

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2022

Mismatch Analysis of DTCs With an Improved BIST-TDC in 28-nm CMOS

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Chen, Peng; Yin, Jun; Zhang, Feifei; Mak, Pui-In; Martins, Rui P.; Staszewski, Robert Bogdan

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A 20 MHz-2 GHz Inductorless Two-Fold Noise-Canceling Low-Noise Amplifier in 28-nm CMOS

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Bozorg, Amir; Staszewski, Robert Bogdan

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2022

A 529- μ W Fractional-N All-Digital PLL Using TDC Gain Auto-Calibration and an Inverse-Class-F DCO in 65-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chen, Peng; Meng, Xi; Yin, Jun; Mak, Pui-In; Martins, Rui P.; Staszewski, Robert Bogdan

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A Broadband Fully Integrated Power Amplifier Using Waveform Shaping Multi-Resonance Harmonic Matching Network

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Nikandish, G. Reza; Nasri, Abbas; Yousefi, Alireza; Zhu, Anding; Staszewski, Robert Bogdan

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Multirate Timestamp Modeling for Ultralow-Jitter Frequency Synthesis: A Tutorial

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Hu, Yizhe; Siriburanon, Teerachot; Staszewski, Robert Bogdan

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Design of High-IF Discrete-Time Receivers for IoT: Demystifying Aliasing Trade-Offs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ferreira, Sandro Binsfeld; Baumgratz, Filipe D.; Bampi, Sergio; Staszewski, Robert Bogdan

被引用次数: 1

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Cryogenic Transistor Confinement Well Simulation through Material and Carrier Transport Decoupling

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

Power, Conor; Staszewski, Robert Bogdan; Blokhina, Elena

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2022

Modelling of Electron Injection and Confinement in Cryogenic 22-nm FD-SOI Quantum Dot Arrays

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

Power, Conor; Andrade-Miceli, Dennis; Bashir, Imran; Asker, Mike; Leipold, Dirk; Staszewski, Robert Bogdan; Blokhina, Elena

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2022

Control of Quantum Systems: Comparison of Different Techniques by the Example of Charge and Spin Semiconductor Qubits

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

Wu, Xutong; Giounanlis, Panagiotis; Staszewski, Robert Bogdan; Blokhina, Elena

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2022

A 38.6-fJ/Conv.-Step Inverter-Based Continuous-Time Bandpass $\Delta\Sigma$ ADC in 28 nm Using Asynchronous SAR Quantizer

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ghaedrahmati, Hanie; Zhou, Jianjun; Staszewski, Robert Bogdan

被引用次数: 5

2021

A Reference-Waveform Oversampling Technique in a Fractional-N ADPLL

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Du, Jianglin; Siriburanon, Teerachot; Hu, Yizhe; Govindaraj, Vivek; Staszewski, Robert Bogdan

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2021

A Distributed Stubs Technique to Mitigate Flicker Noise Upconversion in a mm-Wave Rotary Traveling-Wave Oscillator

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Shehata, Mohamed Atef; Keaveney, Michael; Staszewski, Robert Bogdan

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2021

Unbalanced Power Amplifier: An Architecture for Broadband Back-Off Efficiency Enhancement

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Nikandish, G. Reza; Staszewski, Robert Bogdan; Zhu, Anding

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2021

A 0.02-4.5-GHz LN(T)A in 28-nm CMOS for 5G Exploiting Noise Reduction and Current Reuse

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bozorg, Amir; Staszewski, Robert Bogdan

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2021

A Type-II Phase-Tracking Receiver

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hu, Suoping; Du, Jianglin; Chen, Peng; Nguyen, Hieu Minh; Quinlan, Philip; Siriburanon, Teerachot; Staszewski, Robert Bogdan

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2021

A 0.7-V Sub-mW Type-II Phase-Tracking Bluetooth Low Energy Receiver in 28-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Hu, Suoping; Chen, Peng; Quinlan, Philip; Staszewski, Robert Bogdan

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2021

An Active-Under-Coil RFDAC With Analog Linear Interpolation in 28-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zhang, Feifei; Chen, Peng; Walling, Jeffrey S.; Zhu, Anding; Staszewski, Robert Bogdan

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2021

A Compact Transformer-Based Fractional-N ADPLL in 10-nm FinFET CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Li, Chao-Chieh; Yuan, Min-Shueh; Liao, Chia-Chun; Chang, Chih-Hsien; Lin, Yu-Tso; Tsai, Tsung-Hsien; Huang, Tien-Chien; Liao, Hsien-Yuan; Lu, Chung-Ting; Kuo, Hung-Yi; Ximenes, Augusto Ronchini; Staszewski, Robert Bogdan

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2021

Oscillator Flicker Phase Noise: A Tutorial

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Hu, Yizhe; Siriburanon, Teerachot; Staszewski, Robert Bogdan

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2021

A Fully Integrated GaN Dual-Channel Power Amplifier With Crosstalk Suppression for 5G Massive MIMO Transmitters

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Nikandish, G. Reza; Staszewski, Robert Bogdan; Zhu, Anding

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2021

CMOS charge qubits and qudits: entanglement entropy and mutual information as an optimization method to construct CNOT and SWAP Gates

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Giounanlis, Panagiotis; Wu, Xutong; Sokolov, Andrii; Petropoulos, Nikolaos; Koskin, Eugene; Bashir, Imran; Leipold, Dirk; Staszewski, Robert Bogdan; Blokhina, Elena

被引用次数: 4

2021

Semiconductor Quantum Computing: Toward a CMOS quantum computer on chip

IEEE NANOTECHNOLOGY MAGAZINE

Nikandish, Reza; Blokhina, Elena; Leipold, Dirk; Staszewski, Robert Bogdan

被引用次数: 14

2021

A Charge-Rotating IIR Filter with Linear Interpolation an High Stop-Band Rejection

ESSCIRC 2021 - IEEE 47TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE (ESSCIRC)

Bozorg, Amir; Staszewski, Robert Bogdan

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2021

A 28-GHz Switched-Filter Phase Shifter with Fine Phase-Tuning Capability Using Back-Gate Biasing in 22-nm FD-SOI CMOS

ESSCIRC 2021 - IEEE 47TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE (ESSCIRC)

Kobal, Enis; Siriburanon, Teerachot; Staszewski, Robert Bogdan; Zhu, Anding

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2021

A Mismatch Calibration Technique for SAR ADCs Based on Deterministic Self-Calibration and Stochastic Quantization

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Bagheri, Mojtaba; Schembari, Filippo; Pourmousavian, Naser; Zare-Hoseini, Hashem; Hasko, David; Staszewski, Robert Bogdan

被引用次数: 18

2020

An Event-Driven Quasi-Level-Crossing Delta Modulator Based on Residue Quantization

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Wang, Hongying; Schembari, Filippo; Staszewski, Robert Bogdan

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2020

Passive SC $\Delta\Sigma$ Modulator Based on Pipelined Charge-Sharing Rotation in 28-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Wang, Hongying; Schembari, Filippo; Staszewski, Robert Bogdan

被引用次数: 6

2020

Breaking the Bandwidth Limit: A Review of Broadband Doherty Power Amplifier Design for 5G

IEEE MICROWAVE MAGAZINE

Nikandish, Gholamreza; Staszewski, Robert Bogdan; Zhu, Anding

被引用次数: 77

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CMOS Position-Based Charge Qubits: Theoretical Analysis of Control and Entanglement

IEEE ACCESS

Blokhina, Elena; Giounanlis, Panagiotis; Mitchell, Andrew; Leipold, Dirk R.; Staszewski, Robert Bogdan

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2020

A Single-Pin Antenna Interface RF Front End Using a Single-MOS DCO-PA and a Push-Pull LNA

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Xu, Kai; Yin, Jun; Mak, Pui-In; Staszewski, Robert Bogdan; Martins, Rui P.

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2020

Broadband Fully Integrated GaN Power Amplifier With Minimum-Inductance BPF Matching and Two-Transistor AM-PM Compensation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Nikandish, G. Reza; Staszewski, Robert Bogdan; Zhu, Anding

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2020

A Switched-Capacitor DC-DC Converter Powering an LC Oscillator to Achieve 85% System Peak Power Efficiency and -65dBc Spurious Tones

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Urso, Alessandro; Chen, Yue; Staszewski, Robert Bogdan; Dijkhuis, Johan F.; Stanzione, Stefano; Liu, Yao-Hong; Serdijn, Wouter A.; Babaie, Masoud

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2020

A Time-Domain 147fsrms 2.5-MHz Bandwidth Two-Step Flash-MASH 1-1-1 Time-to-Digital Converter With Third-Order Noise-Shaping and Mismatch Correction

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Wu, Ying; Lu, Ping; Staszewski, Robert Bogdan

被引用次数: 12

2020

A 0.36-V 5-MS/s Time-Mode Flash ADC With Dickson-Charge-Pump-Based Comparators in 28-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Esmailian, Ali; Schembari, Filippo; Staszewski, Robert Bogdan

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2020

An Adaptive-Resolution Quasi-Level-Crossing Delta Modulator With VCO-Based Residue Quantizer

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Wang, Hongying; Viet Nguyen; Schembari, Filippo; Staszewski, Robert Bogdan

被引用次数: 10

2020

A 0.2-V Three-Winding Transformer-Based DCO in 16-nm FinFET CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Li, Chao-Chieh; Yuan, Min-Shueh; Lin, Yu-Tso; Liao, Chia-Chun; Chang, Chih-Hsien; Staszewski, Robert Bogdan

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2020

Design Considerations of the Interpolative Digital Transmitter for Quantization Noise and Replicas Rejection

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Un, Ka-Fai; Zhang, Feifei; Mak, Pui-In; Martins, Rui P.; Zhu, Anding; Staszewski, Robert Bogdan

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2020

Analytical solutions for N interacting electron system confined in graph of coupled electrostatic semiconductor and superconducting quantum dots in tight-binding model

CRYOGENICS

Pomorski, Krzysztof Dominik; Peczkowski, Pawel; Staszewski, Robert Bogdan

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2020

Design of Highly Linear Broadband Continuous Mode GaN MMIC Power Amplifiers for 5G

IEEE ACCESS

Nikandish, Gholamreza; Staszewski, Robert Bogdan; Zhu, Anding

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2019

Modeling of Semiconductor Electrostatic Qubits Realized Through Coupled Quantum Dots

IEEE ACCESS

Giounanlis, Panagiotis; Blokhina, Elena; Pomorski, Krzysztof; Leipold, Dirk R.; Staszewski, Robert Bogdan

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2019

Challenges in On-Chip Antenna Design and Intergration With RF Receiver Front-End Circuitry in Nanoscale CMOS for 5G Communication Systems

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Hedayati, Mahsa Keshavarz; Abdipour, Abdolali; Shirazi, Reza Sarraf; Ammann, Max J.; Johns, Matthias; Cetintepe, Cagri; Staszewski, Robert Bogdan

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2019

Bandwidth Enhancement of GaN MMIC Doherty Power Amplifiers Using Broadband Transformer-Based Load Modulation Network

IEEE ACCESS

Nikandish, Gholamreza; Staszewski, Robert Bogdan; Zhu, Anding

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2019

A 31- μ W, 148-fs Step, 9-bit Capacitor-DAC-Based Constant-Slope Digital-to-Time Converter in 28-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chen, Peng; Zhang, Feifei; Zong, Zhirui; Hu, Suoping; Siriburanon, Teerachot; Staszewski, Robert Bogdan

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2019

A 1-V Bandgap Reference in 7-nm FinFET With a Programmable Temperature Coefficient and Inaccuracy of $\pm 0.2\%$ From -45°C to 125°C

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kamath, Umanath; Cullen, Edward; Yu, Tao; Jennings, John; Wu, Susan; Lim, Peng; Farley, Brendan; Staszewski, Robert Bogdan

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2019

A 0.85mm² 51%-Efficient 11-dBm Compact DCO-DPA in 16-nm FinFET for Sub-Gigahertz IoT TX Using HD2 Self-Suppression and Pulling Mitigation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Xu, Kai; Kuo, Feng-Wei; Chen, Huan-Neng Ron; Cho, Lan-Chou; Jou, Chewn-Pu; Chen, Mark; Staszewski, Robert Bogdan

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2019

A Low-Noise Fractional-N Digital Frequency Synthesizer With Implicit Frequency Tripling for mm-Wave Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zong, Zhirui; Chen, Peng; Staszewski, Robert Bogdan

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2019

Design and Analysis of a DCO-Based Phase-Tracking RF Receiver for IoT Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Liu, Yao-Hong; Purushothaman, Vijaya Kumar; Bachmann, Christian; Staszewski, Robert Bogdan

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2019

A Supply Pushing Reduction Technique for LC Oscillators Based on Ripple Replication and Cancellation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chen, Yue; Liu, Yao-Hong; Zong, Zhirui; Dijkhuis, Johan; Dolmans, Guido; Staszewski, Robert Bogdan; Babaie, Masoud

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2019

Synchronization-Phase Alignment of All-Digital Phase-Locked Loop Chips for a 60-GHz MIMO Transmitter and Evaluation of Phase Noise Effects

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Salarpour, Mahdi; Farzaneh, Forouhar; Staszewski, Robert Bogdan

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2019

Intuitive Understanding of Flicker Noise Reduction via Narrowing of Conduction Angle in Voltage-Biased Oscillators

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Hu, Yizhe; Siriburanon, Teerachot; Staszewski, Robert Bogdan

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2019

Photon Enhanced Interaction and Entanglement in Semiconductor Position-Based Qubits

APPLIED SCIENCES-BASEL

Giounanlis, Panagiotis; Blokhina, Elena; Leipold, Dirk; Staszewski, Robert Bogdan

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2019

The (R)evolution of Distributed Amplifiers

IEEE MICROWAVE MAGAZINE

Nikandish, Gholamreza; Staszewski, Robert Bogdan; Zhu, Anding

被引用次数: 49

2018

All-Digital PLL for Bluetooth Low Energy Using 32.768-kHz Reference Clock and ≤ 0.45 -V Supply

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Chao-Chieh; Yuan, Min-Shueh; Liao, Chia-Chun; Lin, Yu-Tso; Chang, Chih-Hsien; Staszewski, Robert Bogdan

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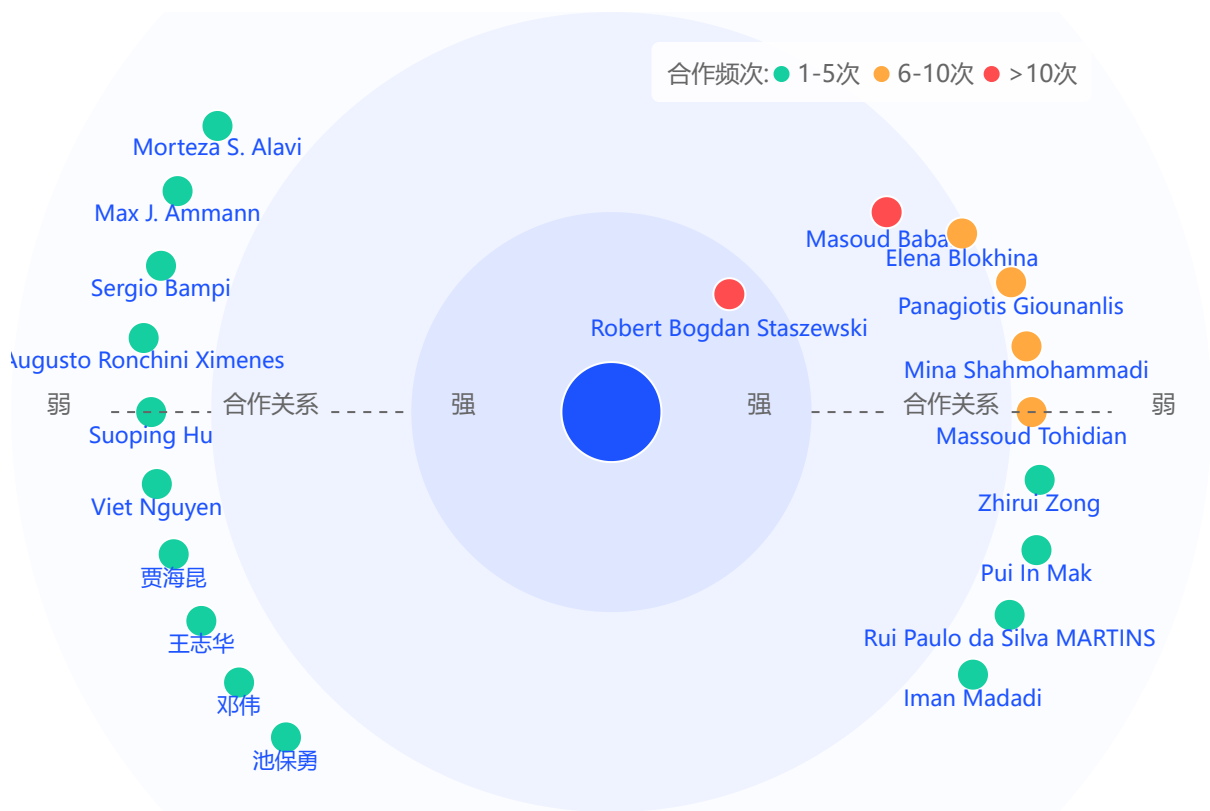
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